

- Brass Column is used to avoid corrosion (or cast iron column with brass lining inside).

Brass / copper trays, highly insulating column.

- $400 \text{ m}^3/\text{h}$ O_2 in India. 56 trays in upper column, 23 trays in lower.

Liq. O_2 is stored in a doubly lined glasswood insulated spherical shell.

* SYNTHESIS GAS :-

CPT :- Petroleum products, Petrochemicals, hydrocarbons, gasoline, petrol etc.

~~First generation Petrochem~~

First generation Petrochemicals $\rightarrow$ obtained directly from petroleum.

① Synthesis gas ^($\text{CO} + \text{H}_2$) is a raw material for 1st gen petrochemicals.

~~It~~ is obtained from steam reforming of hydrocarbons

② Olefins (also, first gen) ^{raw material for} are obtained from pyrolysis of hydrocarbons.

* HEAT RECOVERY

- Product gas exchanges heat with feedstock.

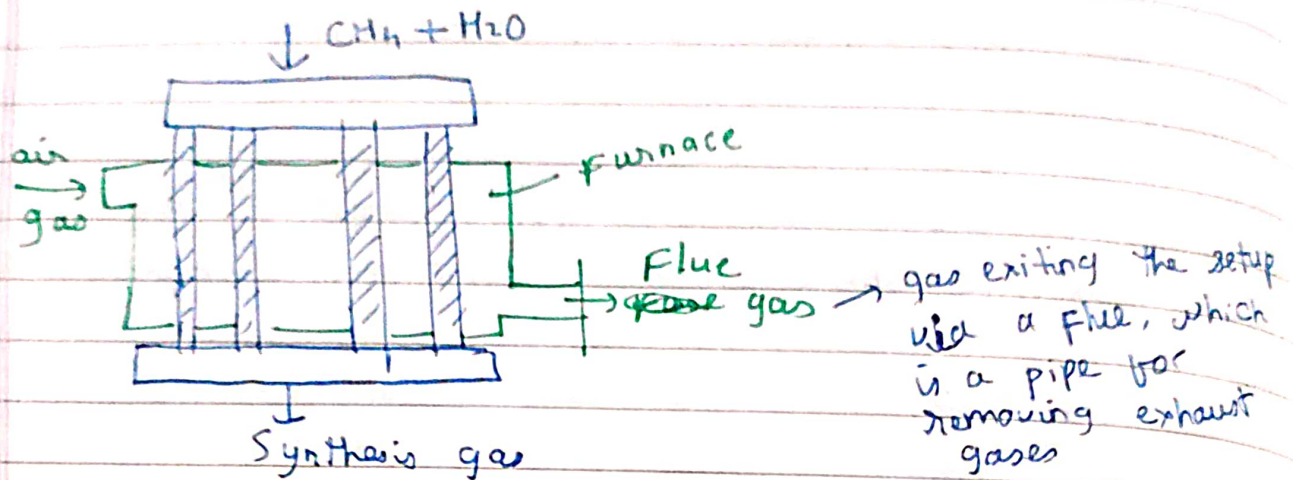
High pressure steam used in SRM etc.

Water Gas

Shift Reaction $\rightarrow \text{CO} + \text{H}_2\text{O} \rightarrow \text{H}_2 + \text{CO}_2$

(Iron chromate catalyst)

* TUBULAR REACTOR FOR STEAM REFORMING

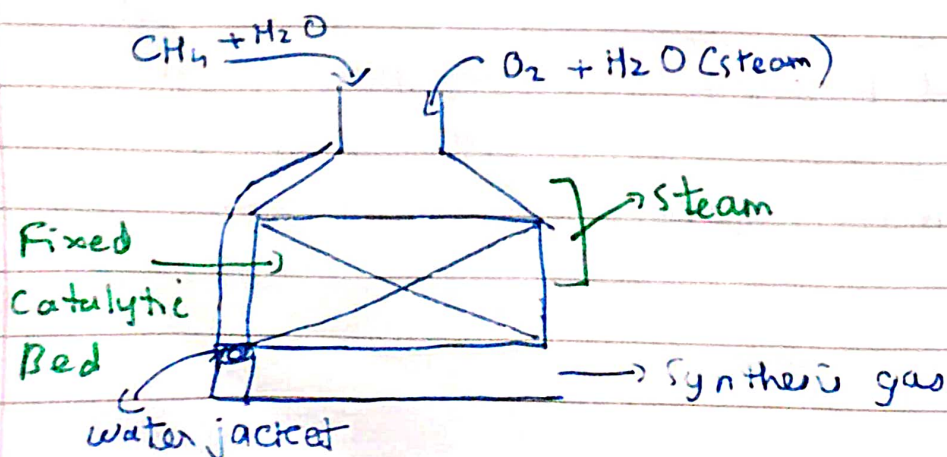


- Sometimes heat adjustment is done by using a part of the feed itself as fuel (by burning it). These types of reactions are called Autothermal Reactions.

Fuel gas to oxygen $\Rightarrow$ 1:0.55

- In an oxidative / auto thermal process, furnace is usually not required. Hydrocarbon: $\text{O}_2 \approx 1:0.55$. A part of H.C. is burnt in the reactor to make up the heat loss by the endothermic S.R. reaction by the endothermic combustion of H.C.

Shaft - continuous Reactor :



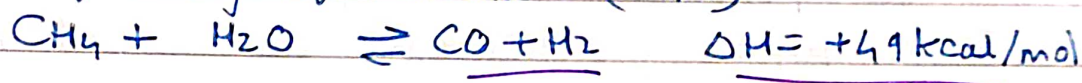
③ Aromatics From Catalytic Reforming of Gasoline

Also raw materials for 1st gen petrochemicals

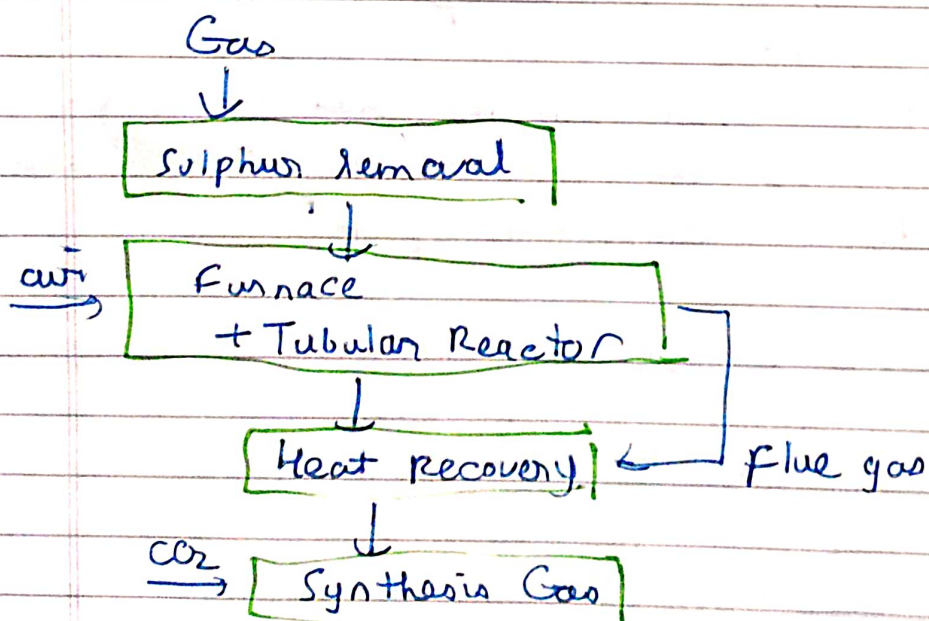
- It is difficult to obtain pure $\text{CO} + \text{H}_2$: some CO_2 is always present. (which is sometimes beneficial too).

H_2 to $(\text{CO} + \text{CO}_2)$ ratio is analysed.
most precious product.

→ Steam Reforming of Methane: (SRM)



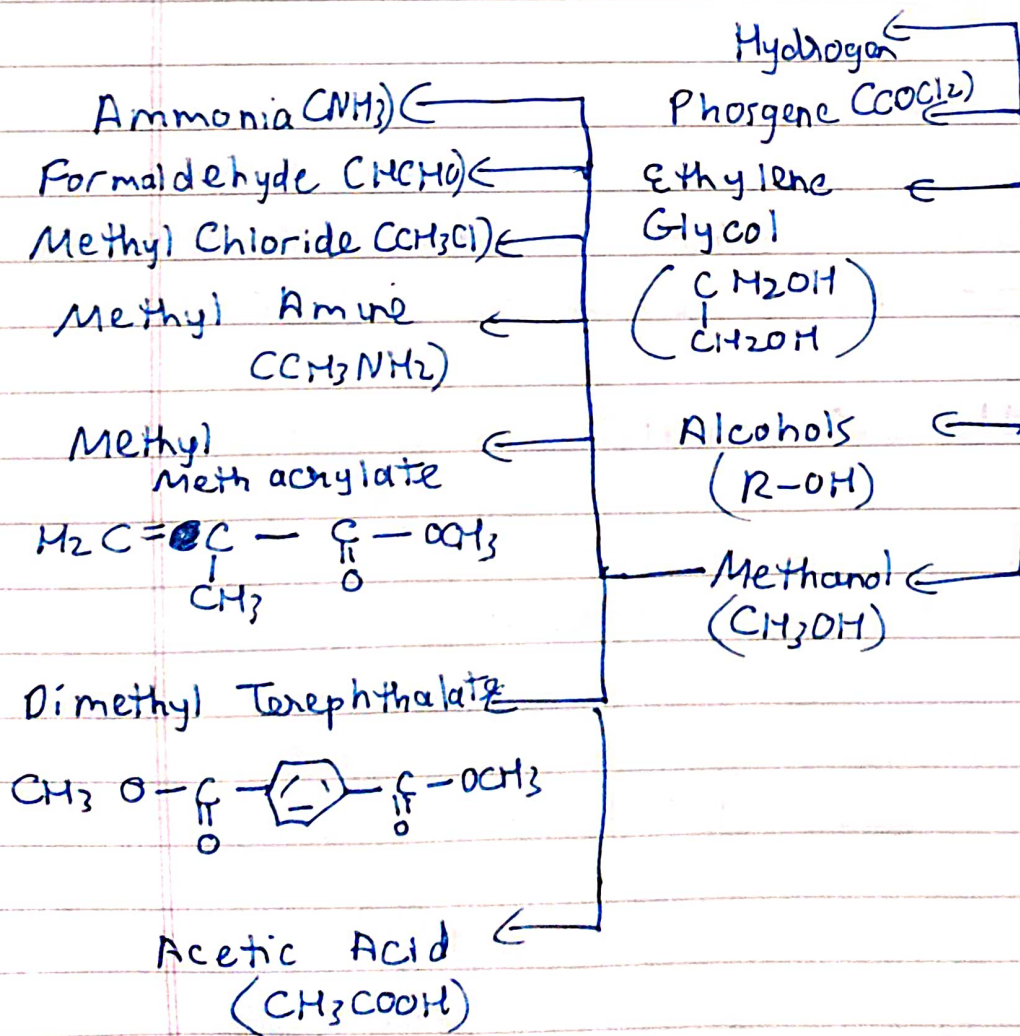
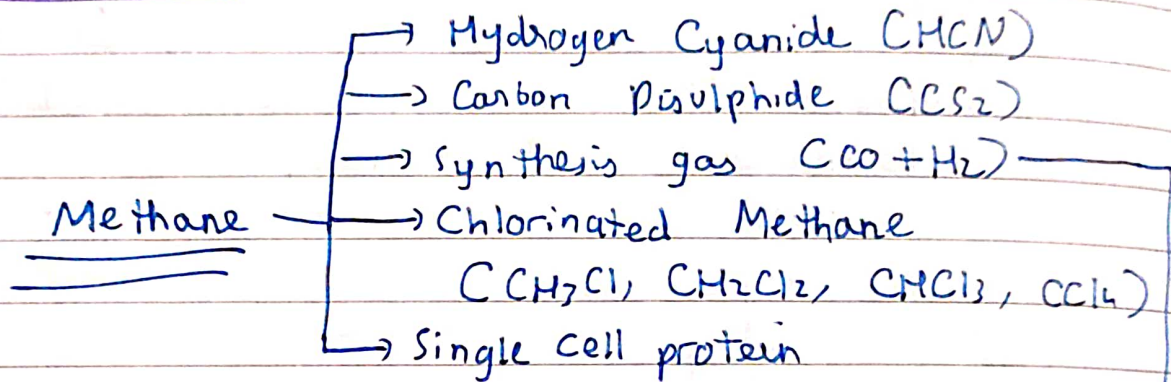
→ Water - Gas shift reaction / shift conversion



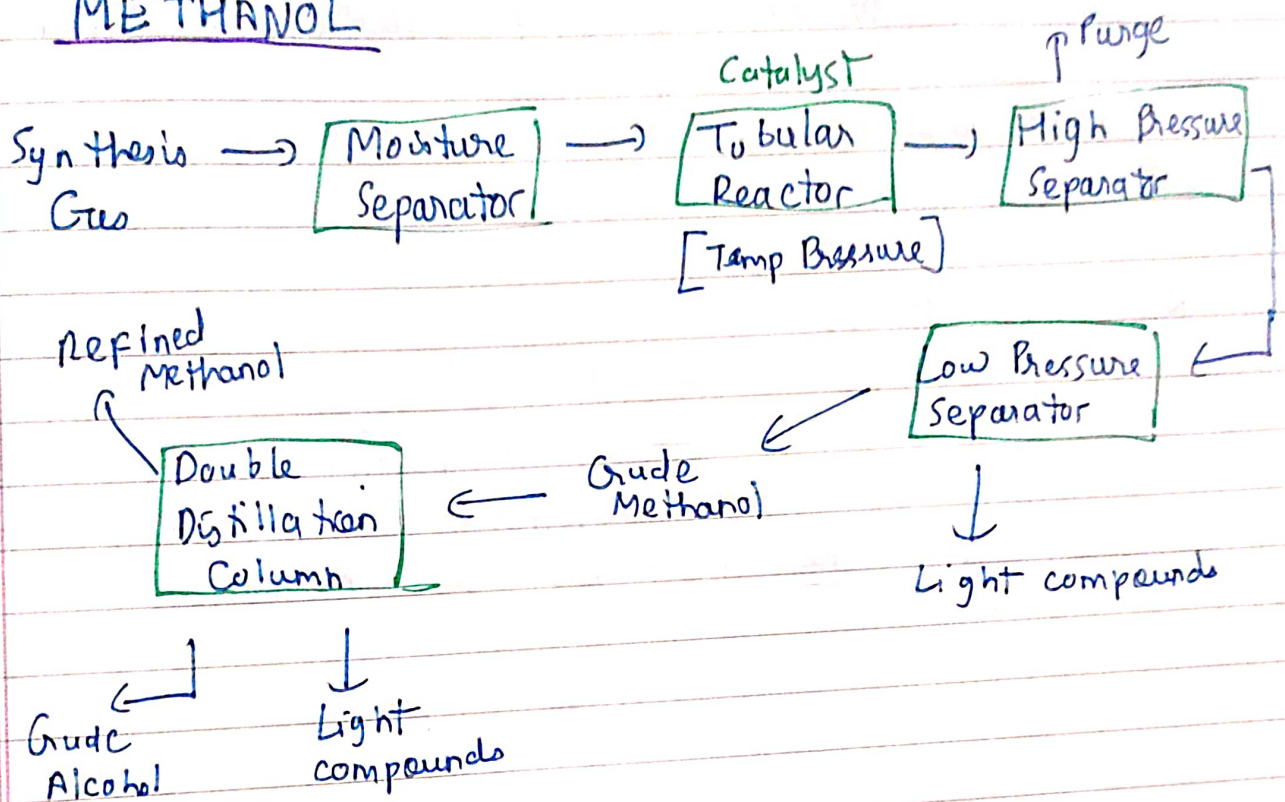
Composition of Synthesis gas from Methane Stream reforming	
Components	Vol %
CO	15.5
CO_2	8.1
H_2	75.7
N_2	0.2
CH_4	0.5
H_2S	~0

Catalyst : $\text{Ni} / \text{Al}_2\text{O}_3 / \text{Ca-Aluminate} / \text{Mg-Aluminate}$
 (6-20% by weight)

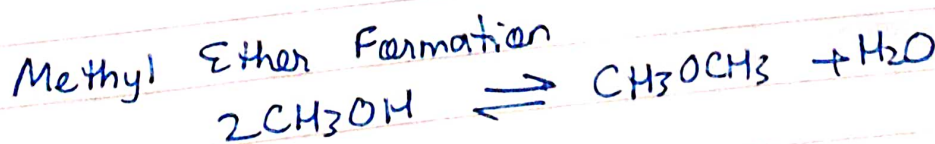
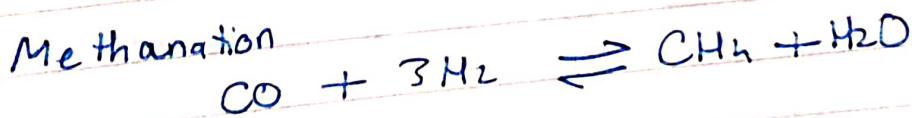
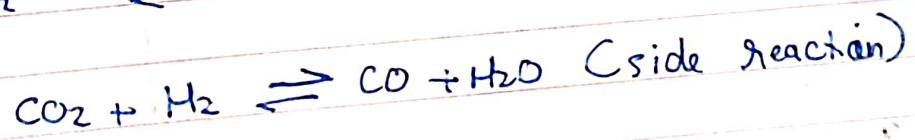
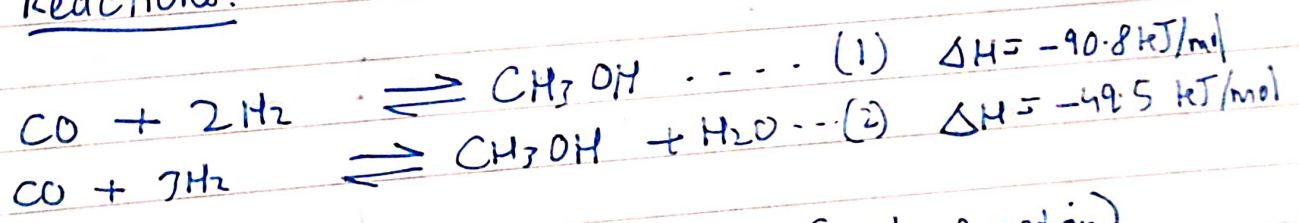
* METHANE



* METHANOL



* Reactions:



Side
rxns to be
avoided

→ Carbon Efficiency / Selectivity of Methanol

$$\hookrightarrow \frac{\text{No. of } \text{CH}_3\text{OH} \text{ moles produced}}{\text{No. of } (\text{CO} + \text{CO}_2) \text{ moles in syn. gas}} \times 100$$

- * Production of Methanol is favoured by:
- Elevation of P
 - Reduction of T
 - Increase of CO/CO₂ ratio
 - Increase of H₂ content in syn. gas

Thorium + Copper $\xrightarrow[\text{merged}]{\text{molten and}}$ ThCu₆ → new catalyst
 Intermetallic

continuous decrease in
 undesired side rxns, and
 production of methanol

10 times more
 active than Cu.

Very good stability. Sustains its
 activity upto 475 hours.

HCHO → used in resins

Methanol is also used in producing anti knock
 agents.

* ETHANOL

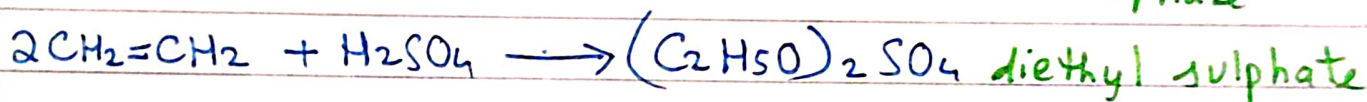
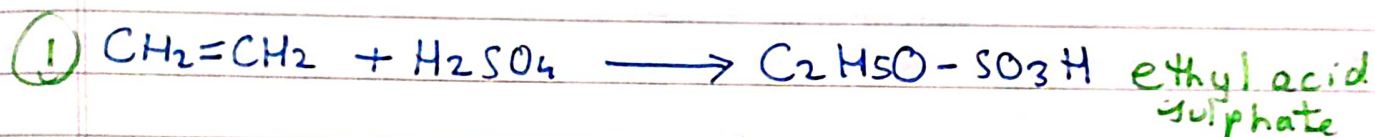
- Used as a solvent.
- Used as a fuel additive ~~into~~ C to gasoline, for raising the vapour of the fluid for spark ignition).
- High ~~octet~~ ~~number~~ octane number (more octane number, superior is the petrol).
- Produced by :-
 - Fermentation
 - Petroleum source, ethylene
 - Indirect and direct hydration

Hence

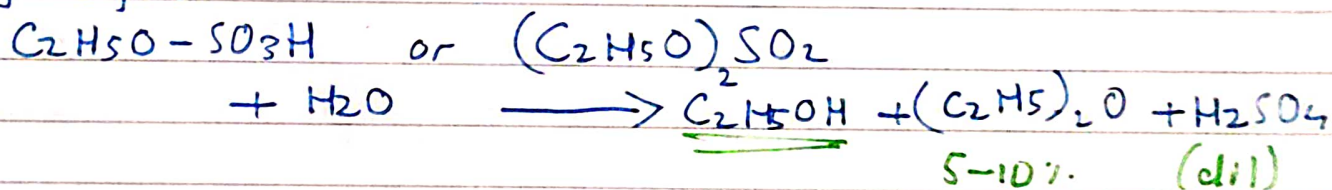
Produced by:

- Fermentation process from agricultural wastes.
- Synthesis from ethylene:-
 - a) Indirect hydration of ethylene
 - b) Direct hydration of ethylene

→ Indirect hydration



(2) Hydrolysis:-



, Reconciliation

Evaporation ← of dil H_2SO_4
under vacuum

Condition of reaction in absorber

Feed → min 35% vol % of ethylene

(methane + ethane)

Counter current separation in 98% H_2SO_4 in absorber

13-15 atm

$1.3 - 1.5 \times 10^6 \text{ Pa}$

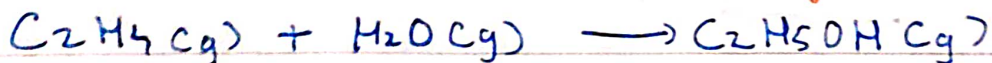
Exothermic
Reaction

- T is controlled below 90°C otherwise resins and oils are formed
- Absorb^{er} is reduced externally to maintain T. Accurate V of reactant is used to keep T between $80-90^{\circ}\text{C}$.
- Ester liquid is removed from absorber and then reacted in another reactor to form $\text{C}_2\text{H}_5\text{OH}$. Stripper separation of final product to remove H_2SO_4 and other impurities.
- Corrosiveness of conc. H_2SO_4 .
Production of good amount of diethyl ether. highly volatile mixes with ethanol
Costly process
which is why this is not the most popular process.

→ Direct Hydration.

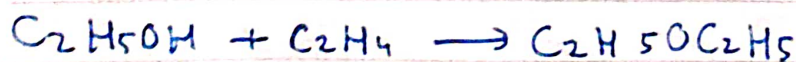
- Catalytic reaction, exothermic

Catalytic Hydration Process



$$\Delta = -43.4 \text{ kJ}$$

- Side Reaction $\rightarrow$ unwanted happens at lower T)



- At high P, ethylene polymerises

- Molar ratio of ethylene : de-ionised water

$$\Rightarrow 1:0.3 \text{ to } 1:0.8$$

$$T \rightarrow 250-300^{\circ}\text{C}$$

$$P \rightarrow 6-8 \text{ MPa}$$

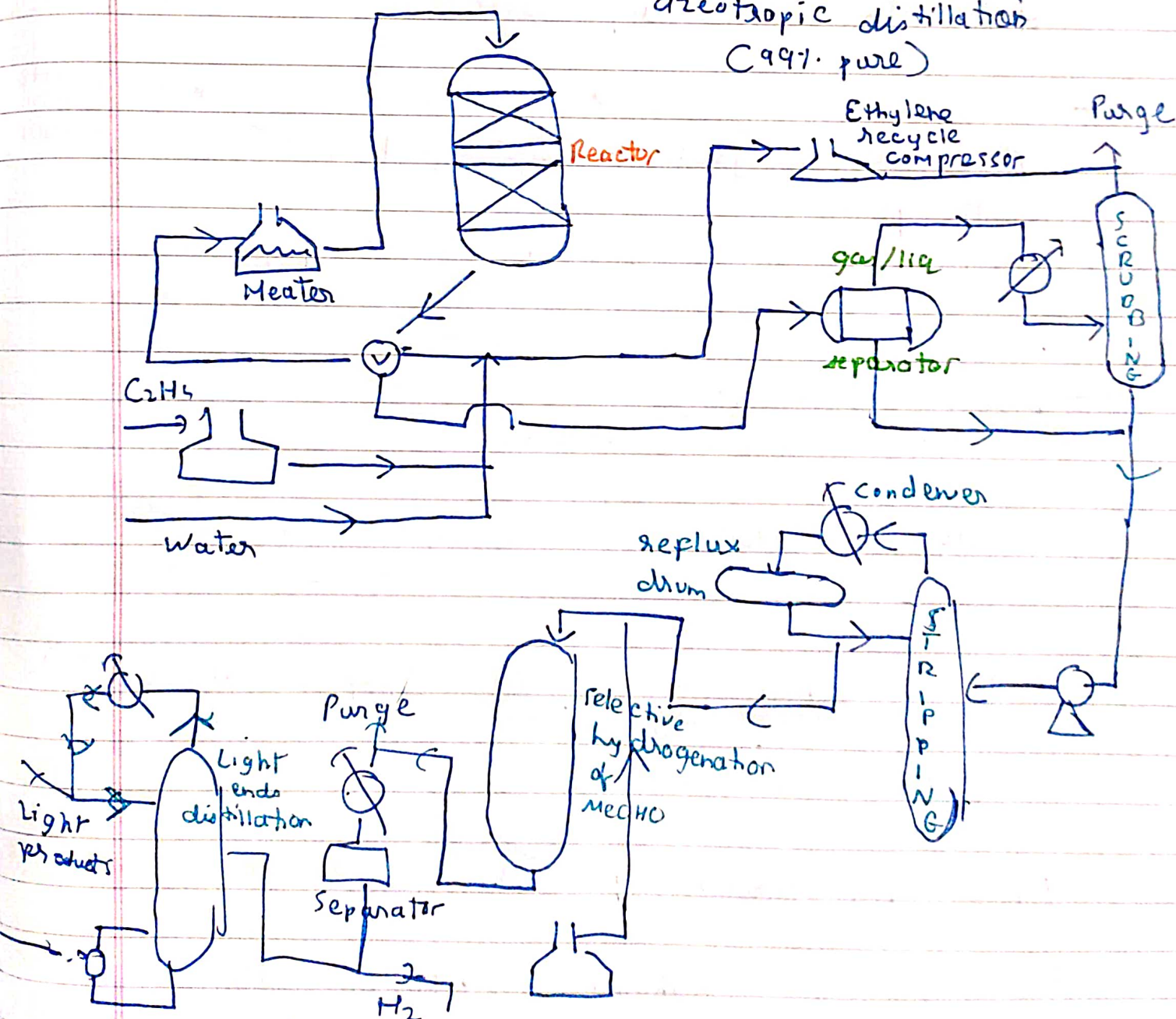
Catalyst: Phosphoric Acid on silica

When feed vapour passes through catalysts, it might pick phosphoric acid with it. To prevent that, we introduce NaOH to neutralize the stream.

First distillation Column $\rightarrow$ Small impurities and Ether removed.

Second distillation Column $\rightarrow$ Water - Ethanol mixture (Azeotrope, 90-95% purity).

$\downarrow$ to remove this azeotropic distillation (99% pure)



Azeotropic distillation column for azeotropes
High reflux ratio and large number of trays.

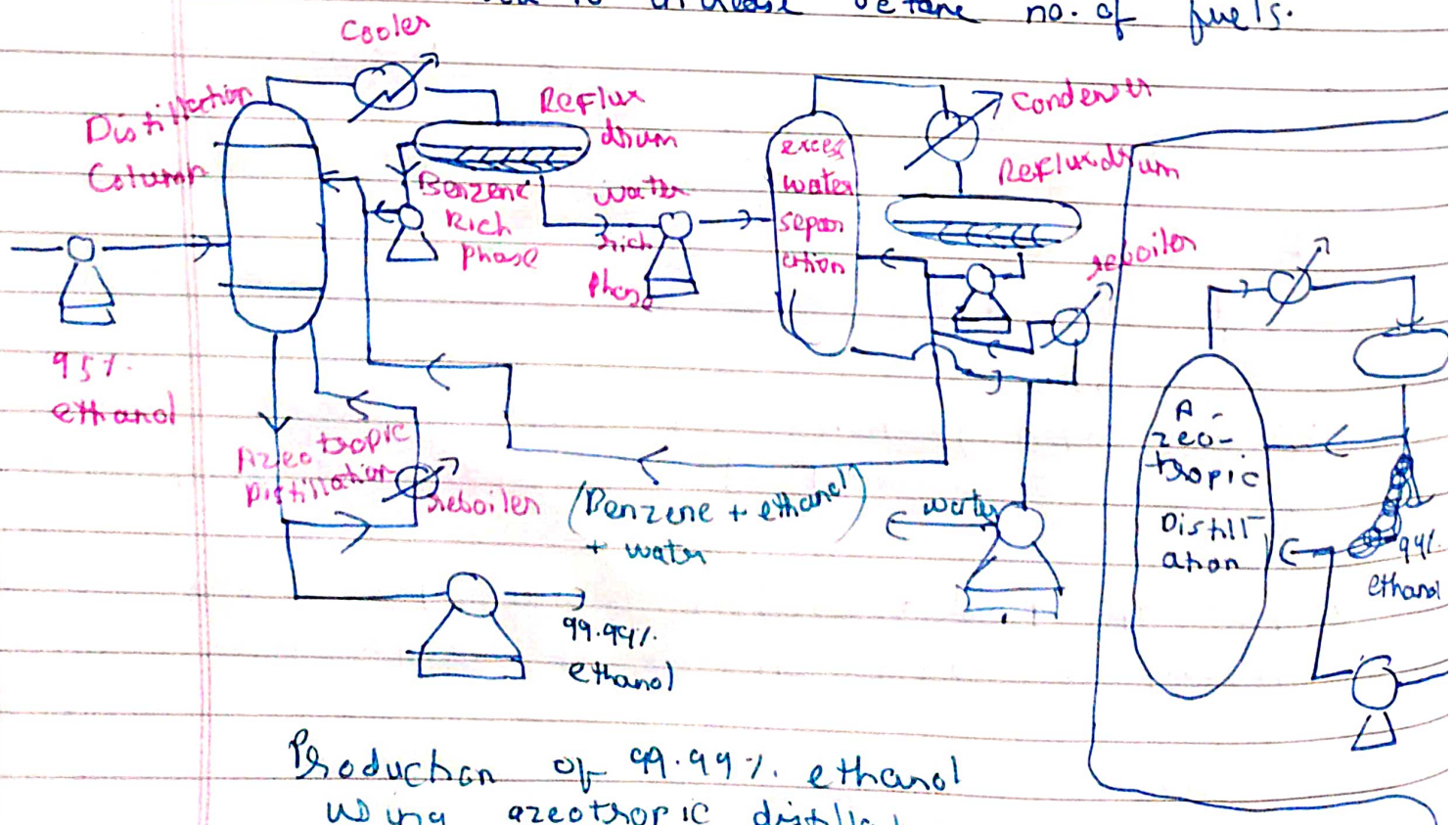
When acetaldehyde (MeCHO) is formed, nickel bed is used and hydrogen gas is passed through.

- Below 12 kPa pressure, ethanol-water azeotrope breaks down. At 95% ethanol-water, same boiling point in liq-vapour phases is observed $\rightarrow$ Azeotropic distillation.

- From top of azeotropic distillation, ternary mixture entrainers $\leftarrow$ Benzene, ethanol, water

Solvent will contain the heavier liquid (contains the lighter liquid in this case).

Ethanol $\rightarrow$ used to increase octane no. of fuels.



Production of 99.99% ethanol using azeotropic distillation

* HIGHER ALCOHOLS

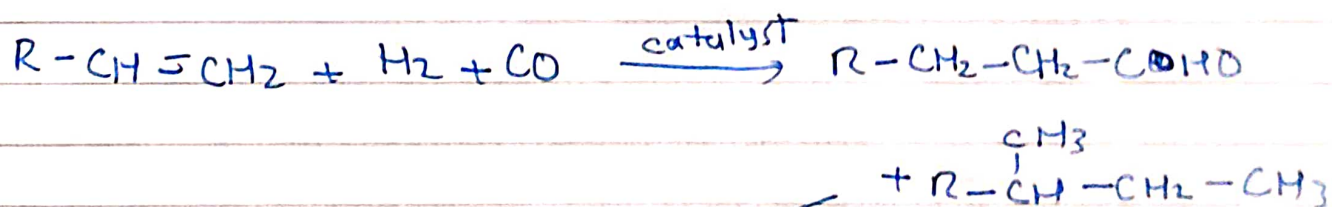
- $C_3 - C_{20} \rightarrow$ have many uses
- Prepared from synthesis gas - reaction with olefin
 $\downarrow$
 produces aldehyde with 1 carbon more than olefin
 $\leftarrow$ hydrogenated to reduce it to an alcohol

Oxo Process:- Followed by hydroformylation

$\rightarrow$ low conversion, low selectivity

Initially $Co(CO)_n$ was used, but now replaced by $Rh-(CO)_n$ modified by Phosphine.

$\rightarrow$ very sensitive to poisoning



$\swarrow$
Further reduced to alcohol

$P < 20 \text{ bar}$

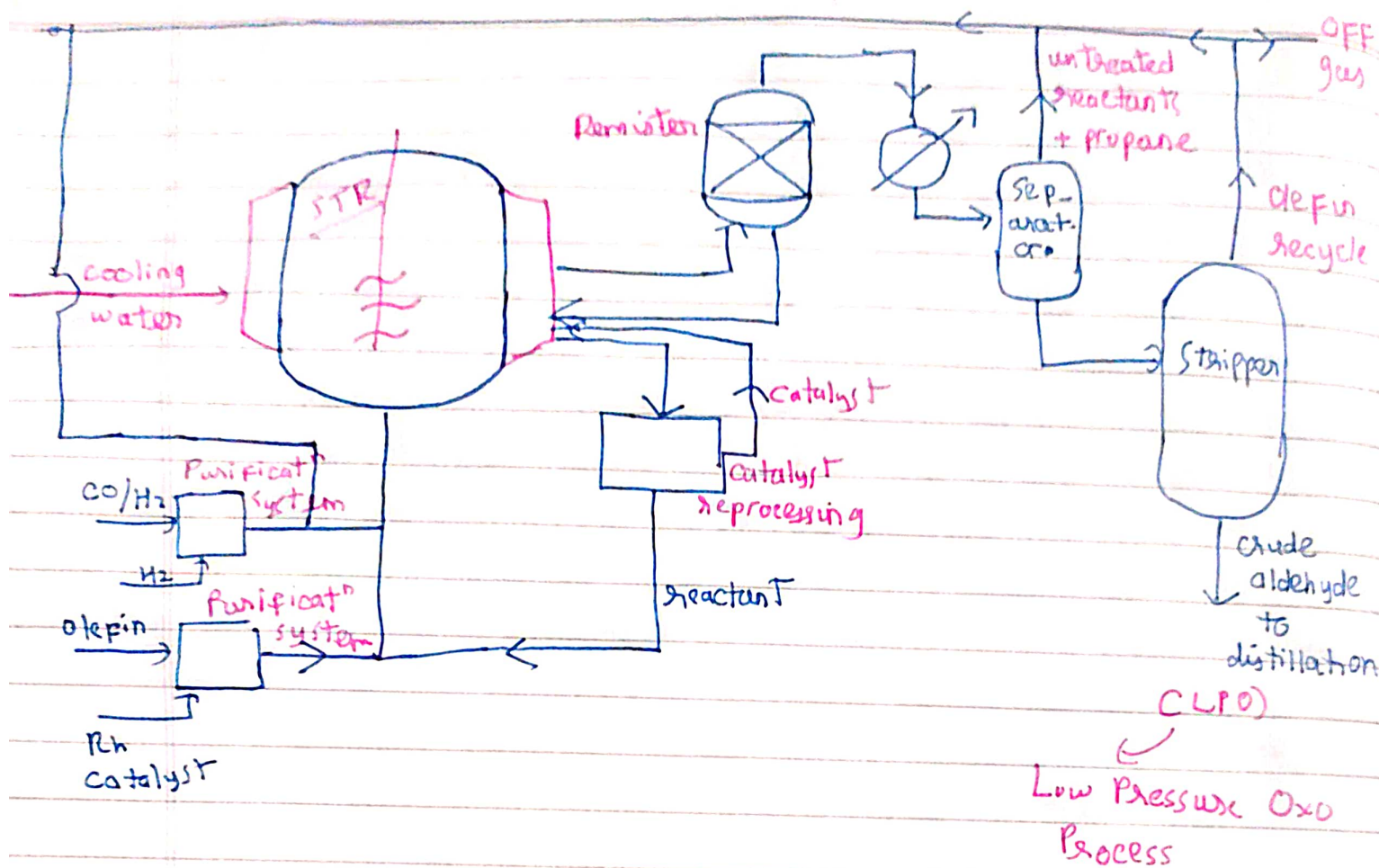
$T \rightarrow 35 - 115^\circ C$

Per pass conversion $\rightarrow 70\%$

Water ~~cooler~~ ^{cooler} jacketed by batch reactor

$\rightarrow$ Feed introduced after purification.

A part of Feed + Catalyst is sent for poison removing, to reduce chances of deactivation.



Demister $\rightarrow$ removes fine catalyst particles and subside vapour.

Oxidation of Paraffin to higher alcohols: Not frequently used as it is not selective.

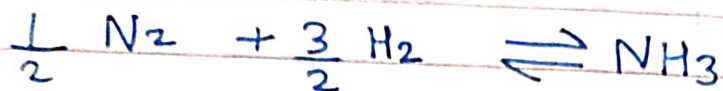
Boric acid $\rightarrow$ Borate Salts
 $\downarrow$
 hydrolysed

Paraffin + air	$\xrightarrow[0.1 \text{ wt} \% \text{ KMnO}_4]{150-200^\circ\text{C}}$	70 mol % alcohol
	+ 5 wt % boric acid	20 mol % ketones
		10 mol % organic acids

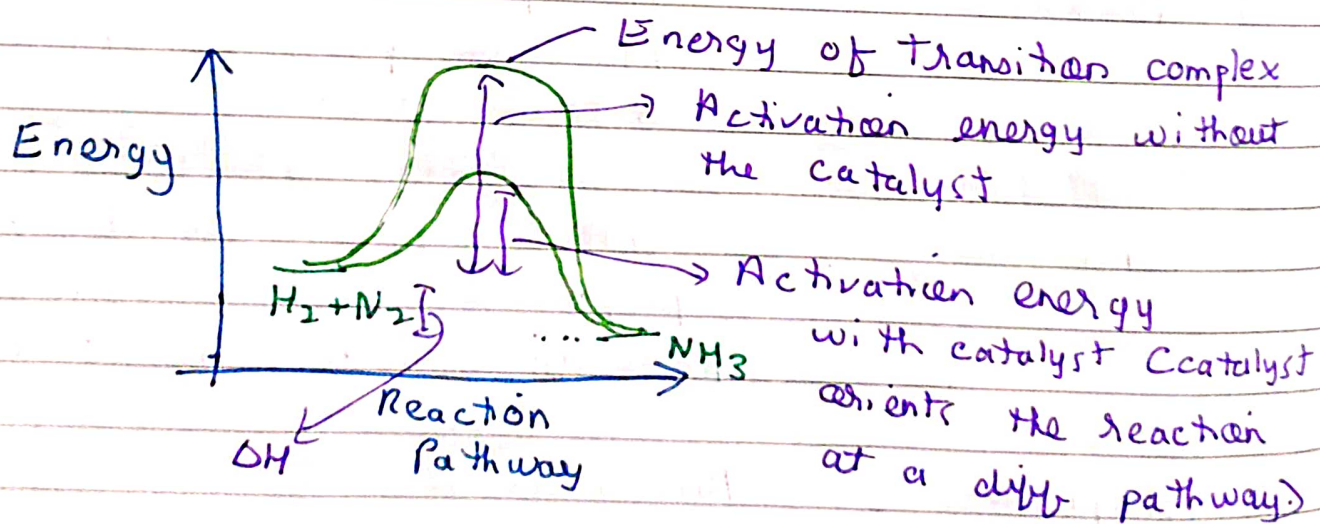
- Unconverted Paraffin obtained by distillation

* PREPARATION OF AMMONIA (NH₃)

Haber's Process:



- Increasing the pressure leads to more production of ammonia.
- Exothermic rxn $\Rightarrow$ Low temp favours the rxn.
- However, you can't decrease T to ambient temp because N-N bonds ~~have~~ are really strong at atmospheric pressure.
- Haber used 500°C and 150-200 atm, however the yield of ammonia was very low.
- Initial catalysts were Os and U₃O₈ but they were costly.
- Finally, Fe₂O₃ was used as the catalyst.



$$K_{eq} = \frac{P_{\text{NH}_3}}{(P_{\text{N}_2})^{1/2} (P_{\text{H}_2})^{3/2}}$$

- Haber also recycled the reactants, and the overall conversion increased.

- Catalyst: Iron oxide, with oxides of Aluminium, Zirconium or Silicon at about 3%.
- Al_2O_3 ZrO_2 SiO_2
- + potassium oxide at about 10%.

- ^{Space} ~~Stage~~ velocity: Vol of reactants per unit vol of catalyst per unit time
- We change space velocity to increase yield.

$$\text{Space Time} = \frac{1}{\text{Space velocity}}$$

* USES OF AMMONIA

<u>Industry</u>	<u>Use</u>
1. Fertilizer	For production of: Ammonium Sulphate, Ammonium phosphate, Ammonium Nitrate, Urea.
2. Chemicals	Synthesis of: Nitric Acid, (for preparation of T.N.T), nitroglycerin, hydrogen cyanide, Hydrazine
3. Explosives	Ammonium Nitrate
4. Fibre and Plastics.	Nylon and other Polyamides
5. Refrigeration	Used for making ice, large scale refrigeration plant, air conditioning units

6. Pharmaceuticals Manufacturing of drugs: Sulfonamide
Anti-malaria drugs.
Manufacturing of vitamin B: ~~B₁₂~~
Nicotinamide, Thiamine
7. Pulp and Paper Ammonium bisulphate
8. Mining and Metallurgy Nitriding (bright annealing) of steel, used in zinc and nickel extraction.

- 10,000 to 20,000 / m³ vol of reactant per vol catalyst used in ammonia plant.
- 8 to 15% conversion per pass.
- Triply promoted catalyst, 300 atm, 400-500°C.
- We can reduce the pressure from 300 atm to as low as 400 atm by using Ru/graphite instead of iron oxide.

* AMMONIA PRODUCTION

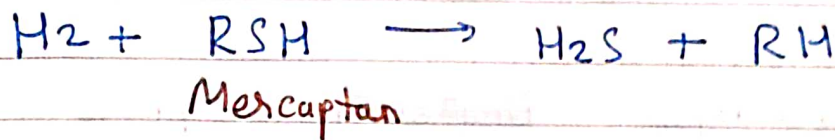
Raw materials for syn gas / steam reforming process → CH₄, LPG, liq hydrocarbons.

$\text{H}_2\text{S} + \text{Zn} \rightarrow \text{ZnS}$ ⇒ done to remove S from Hydrocarbons

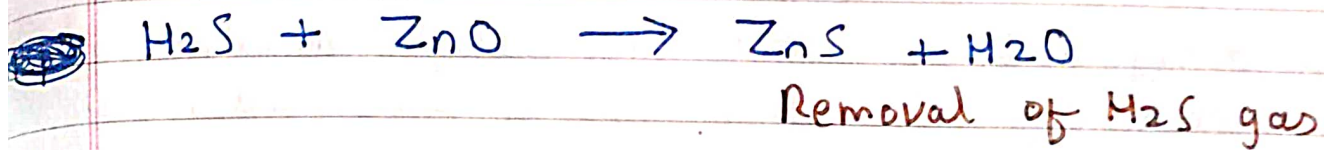
$\text{S} + \text{H}_2 \rightarrow$ hydrodesulphurization process.

* Natural gas (C₄H₁₀) Feed Stock:-

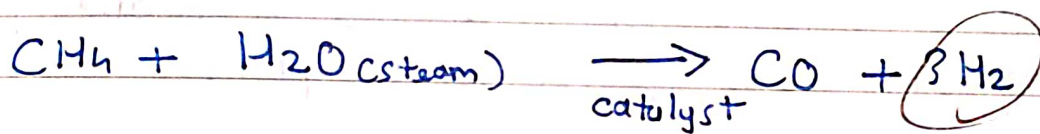
Step 1:- Sulphur Removal from Feed stock



R- Alkyl group.



Step 2:- Steam Reforming Process

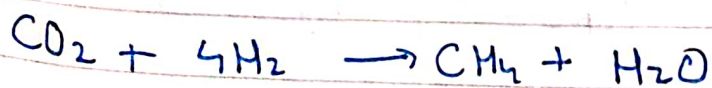


Step 3:- Shift Conversion $\rightarrow$ to convert more CO



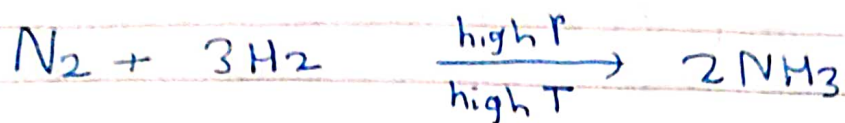
$\rightarrow$ CO₂ is removed either by absorption in ethanol or amine solution OR by adsorption by Pressure-swing adsorption.

Step 4:- Converting ^{remaining} CO to Methane and CO₂



} Methanation Rⁿ to remove any residual CO or CO₂.

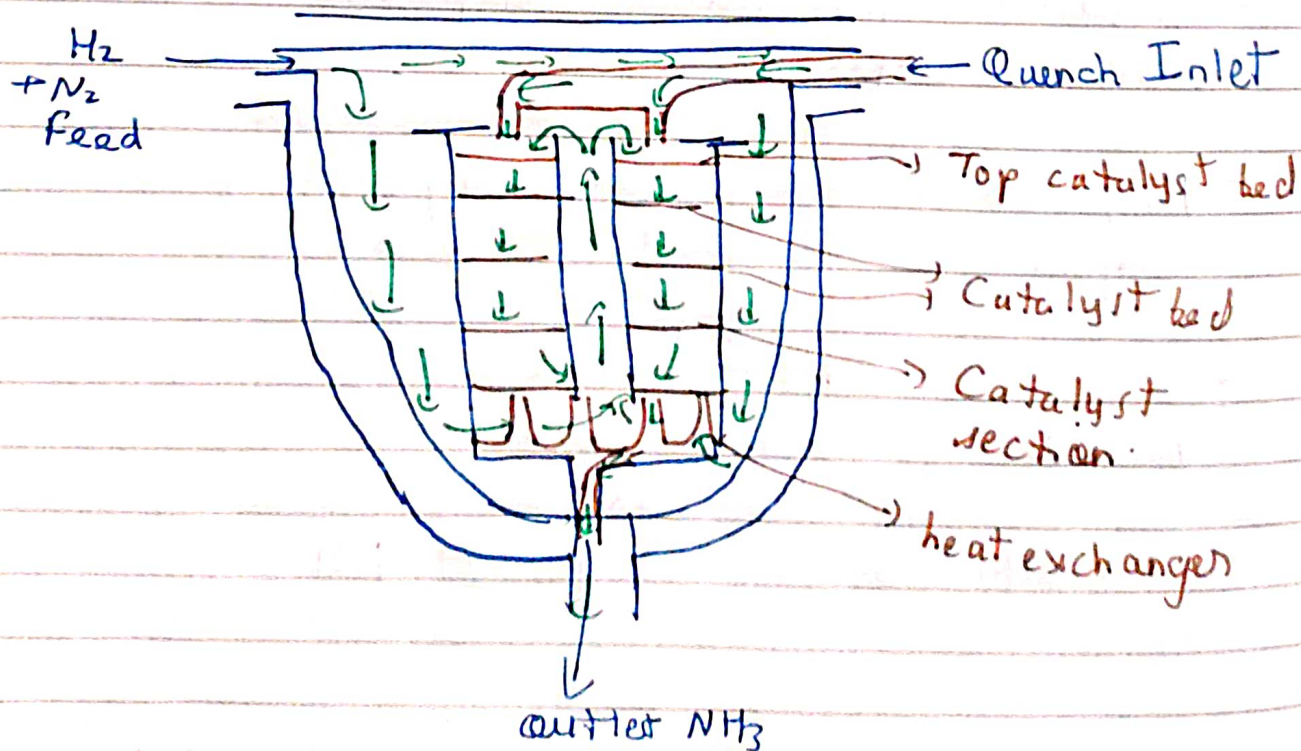
Finally,



Haber - Bosch process

- All these processes operate within 25-35 bar
↓
Step 1 to 4.
- Haber Bosch process operates at 60-180 bar.
licensed
- Some famous designers for Haber's process
eg) Denmark → Halder - Topsøe
Uhde GMBH → Germany
USA → Kellogg and Brown

AMMONIA CONVERTER



Heat exchanger exchanges heat between Feed and high temp Ammonia.

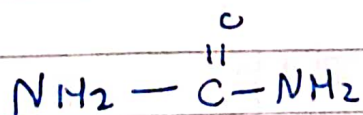
Catalyst screens from top to bottom $\rightarrow$ size increases.

Quench feed (cold) mixes with the hot feed $\rightarrow$ 370-425°C
to maintain temp to maintain reaction temp.

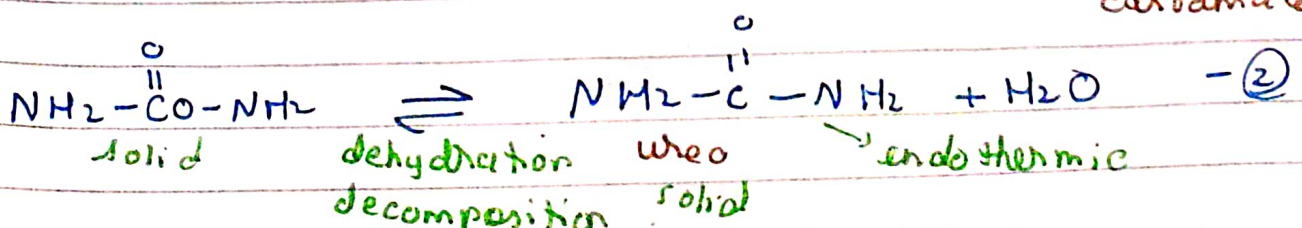
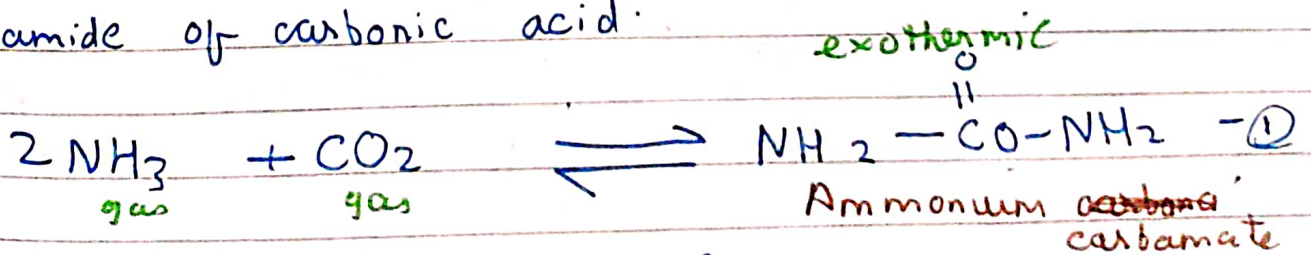
Unreacted H_2 and N_2 is recycled at each stage.

* UREA

- Important fertilizer, solid. Nitrogenous based. Convenient to carry, as compared to N_2 fertilizers.
- It liberates nitrogenous compounds which enhances soil for crops.
- Preparation of organic compounds from 2 inorganic compounds. $(NH_3 + CO_2) \rightarrow$ (urea)



Diamide of carbonic acid.



- colourless, tasteless, odourless.
- urea, when sprayed, decomposes into ammonium carbamate, and if it rains or there is moisture, it further decomposes into NH_3 and CO_2 , which is a bad plants.

① $\rightarrow 150-200^\circ\text{C}$
 $137-232 \text{ MPa}$

* PREP^N OF SOLID UREA $\rightarrow$ from molten urea

1. Drilling $\begin{cases} \text{a) fluidised} \\ \text{b) Non fluidised} \end{cases}$
2. Granulating $\begin{cases} \text{a) Drum granulation} \\ \text{b) Pan granulation} \end{cases}$

1. Molten urea is sprayed from top of tower, and then beads of urea is formed due to strong counter current of air.

2a) Drum Granulation: Better process for uniform size granules. Drum $\rightarrow$ 14 feet diameter. Urea seeds are used. Molten urea ~~deposits~~ ^{deposits} on the urea seeds, similar to the crystallization process (nucleation).

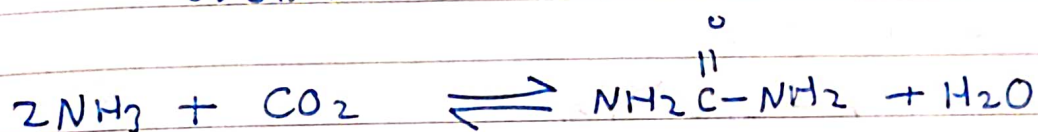
b) Pan Granulation: Same as drum, but a big pan is used.

Granulation $>$ Drilling.

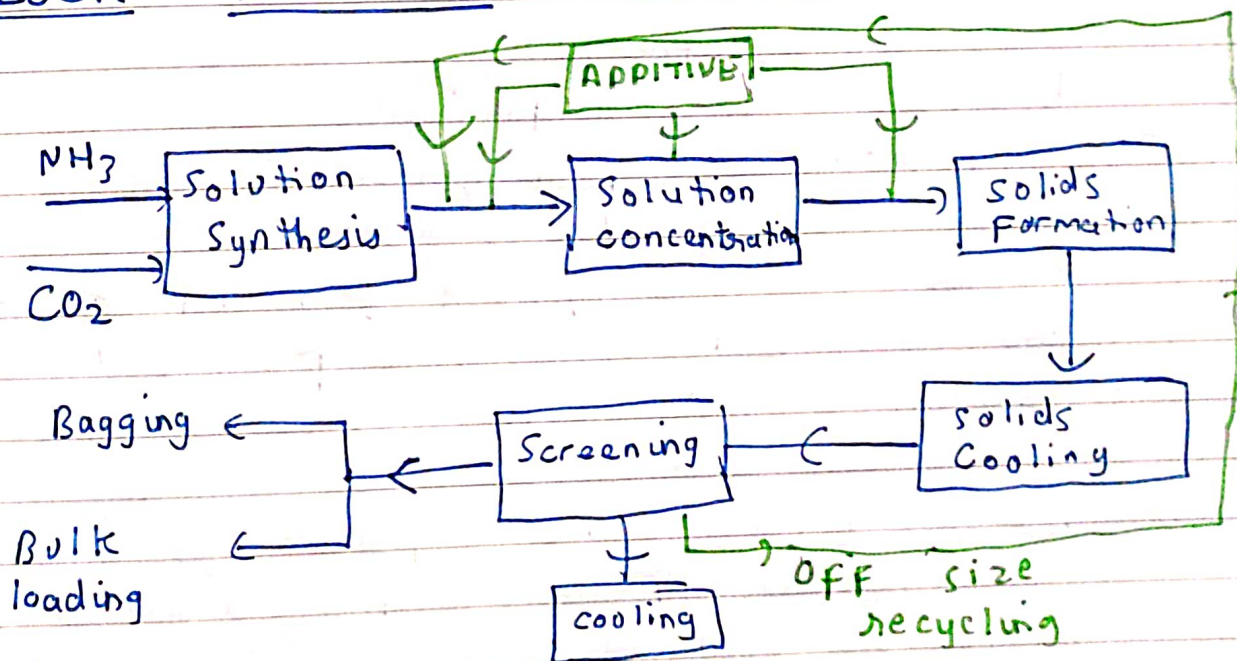
- To increase yield of urea $> 53\%$, you can increase temp, but not beyond 200°C . Also, you can use more ammonia.

We get CO_2 from Ammonia synthesis plant as well. $\rightarrow$ Byproduct stream, ^{vented} ~~taken~~ from carbon ^{dioxide} removal unit.

Overall rxn



* BLOCK DIAGRAM



MAJOR UREA MANUFACTURING OPERATIONS

- Mole ratio of $\text{NH}_3:\text{CO}_2 > 2.5:1$. Depending upon the Feed mole ratio, more or less carbamate is converted to urea per pass through the reactor.
- The P of the first rxn should either be lesser ^{Block Flow} or ~~per~~ P should ^{Flow} be $> \text{VP}$ of Ammonium carbamate. _{Reaction}
- $T_{\text{max}} \approx 185^\circ\text{C}$ ~~as~~ as there are both exo and endothermic

* ONCE THROUGH UREA PROCESS:-

- Single pass. No recycling. There are unreacted NH_3 and Ammonium carbamate, which is passed through dil acids (H_2SO_4 , HCl , HNO_3) to convert them into respective ammonium salts.
- We lose a lot of ammonia which could have been used to make urea. Hence this process is less popular.

* SOLUTION RECYCLE PROCESS:-

- Here the reactor effluent (NH_3 and CO_2 and ammonium carbamate) are absorbed in H_2O to form aqueous ammonium carbamate, which is then recycled.

* UREA (DSM) PROCESS

↳ Stamicarbon BV Process

- Reactor Pressure : 130-150 atm
- "—" Temp : 170-~~180~~ 190°C
- $\text{NH}_3 : \text{CO}_2 \Rightarrow$ ~~2.8:2.9~~ 2.8:2.9
- Unreacted reactor effluents goes into stripping unit where it is stripped and then condensed to recycle again to reactor.

* WATER CONDITIONING / WATER TREATMENT :-

- Quality and quantity of water is very important for daily lives.

- 2 types of Water sources

Surface Water
(Ponds, Rivers etc)

Ground Water

(Underground)

Harder than surface water.

Cooler than surface water
Used for Chemical industry applications.

- Unlike surface water, ground water contains a lot of minerals etc dissolved in them.

- Ground water is usually called hard water, although surface water can be hard as well.

Ground water contains → dissolved salts of Ca and Mg

bicarbonate, chloride, sulphates, nitrates etc

- Hardness: Dissolved Calcium and Magnesium salts as Calcium Carbonate Equivalent.

→ can be removed by heating

Hardness → Carbonate or Temporary hardness (bicarbonate salts of Ca/Mg)

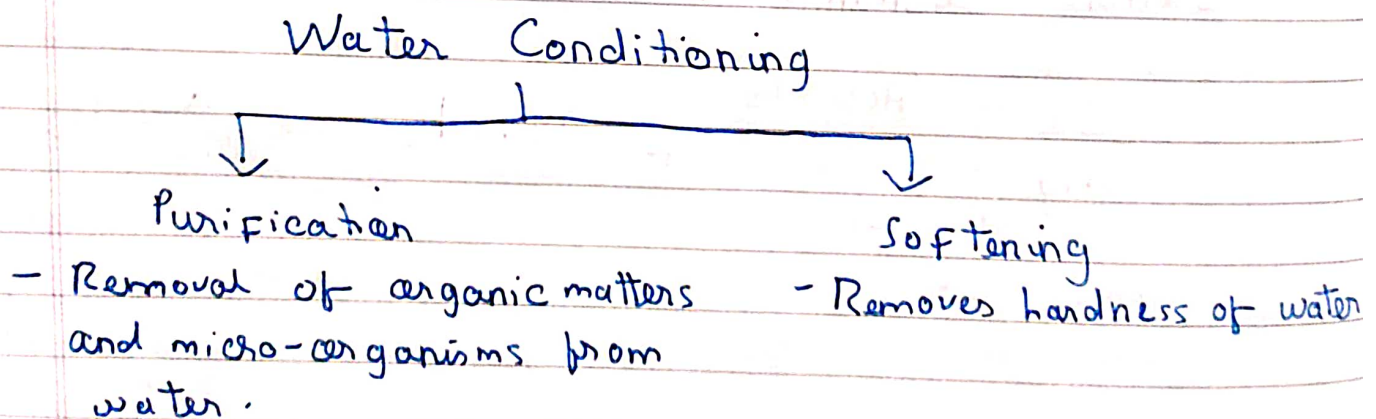
→ Non carbonate or Permanent hardness

- Lec 10
Date _____
Page _____
- Permanent hardness is due to ~~solph~~ sulphates, chlorides of Ca/Mg. Require chemical treatment to remove.

Other Impurities

- TDS $\rightarrow$ Total Dissolved Solids
- Iron
- Colouring agents (you pass water through charcoal to remove these)
- Gases
- ~~H₂S~~ H₂S from hot springs

- * Units of Expressing Water Analysis: ① Parts per million (ppm)
- $\hookrightarrow$ In 1 L, how many mg are dissolved.
- ② grains per gallon
- ③ mg/L



* WATER SOFTENING

- 1) Ion Exchange :- Chemical process in which the mobile hydrated ions from the ion exchanger (ion exchange resin/zeolite), exchange the hard ions from hard water.

It is of 2 types

Cation Exchange

Anion Exchange

↓
- Na-Styrene Divinyl Benzene Sulfonate (Na-resin)

↓
Demineeralization

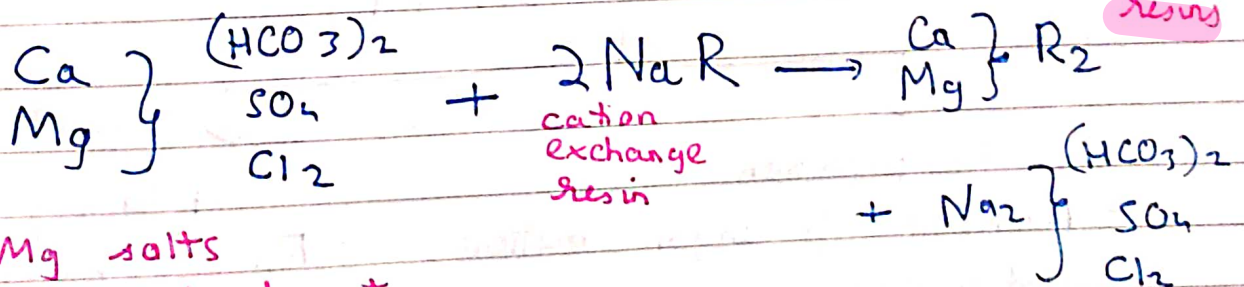
- ~~Na~~ zeolite (sodium)

Alumino silicate $\text{NaAlSi}_3\text{O}_8$

unsure

- In cation exchange Na^+ ion exchanges with $\text{Mg}^{+2}/\text{Ca}^{+2}$ ions.
- Sometimes H^+ ions are also used and we would then get acidic water.
- Condition: $\sim 150^\circ\text{C}$ $\text{pH} \rightarrow 0-14$. Resistant to oxidising agents.

Cation Exchange Reaction:-



Ca/Mg salts dissolved in hard water

cation exchange resin

Insoluble Ca/Mg Resins

sodium salts in water

- Regeneration of resin $\rightarrow$ by NaCl solⁿ.

ANION EXCHANGE $\rightarrow$ WATER $\rightarrow$

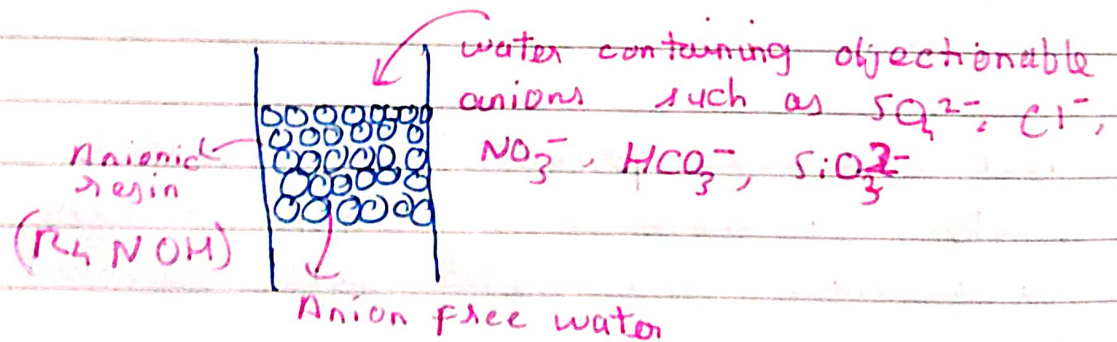
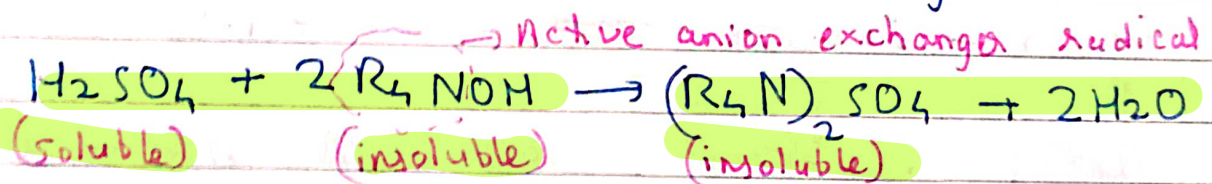
Two resin types

Highly basic

Weakly basic

- For strong acidic ions, Cl^- , SO_4^{2-} , NO_3^- , both resins are useful.
- But for weakly acidic ones, HCO_3^{2-} , silicates etc, highly basic anionic resins are required.

Reaction of ANION removal by Anion Exchanger



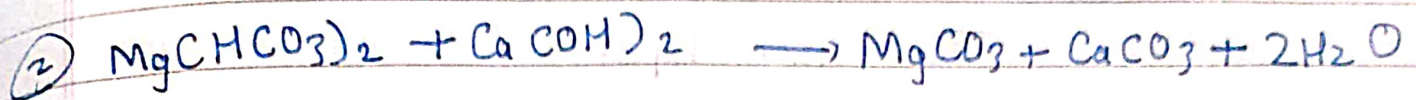
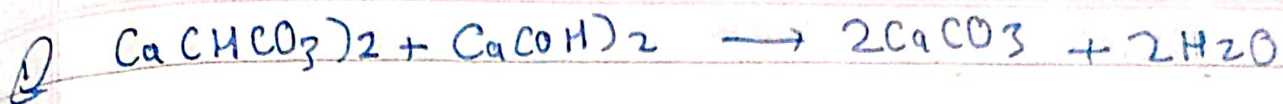
- Highly basic resins are regenerated by caustic soda, while weakly basic resins are regenerated by either NaOH / NH_4OH

$\rightarrow \text{CaCO}_3 \rightarrow \text{Na}_2\text{CO}_3$
 $\rightarrow$ Slaked lime, soda ash

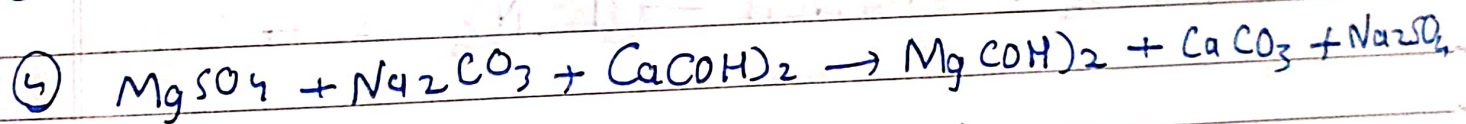
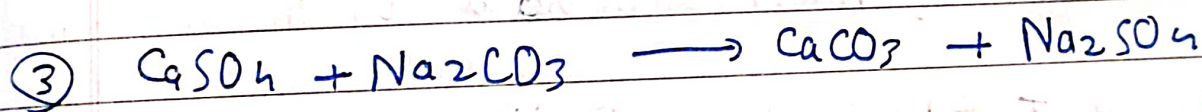
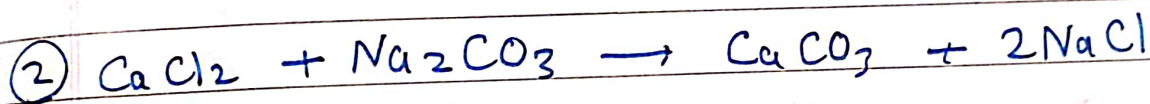
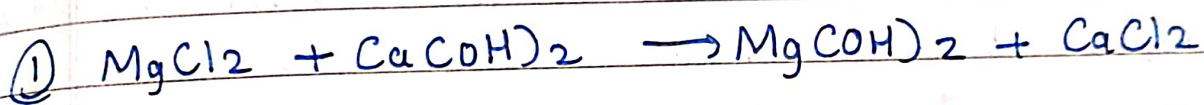
* LIME - SODA PROCESS $\rightarrow$ For removing Ca^{+2} and Mg^{+2} ions

Cold lime soda Process Hot lime soda Process

For carbonate hardness



For non carbonate hardness



- Hot lime soda process is operated at B.P. of water. So, rate of reaction is high also CO_2 is dissolved and water gets removed.

* Silica in waste water is removed by dolomite or activated Magnesia in the softener

* DEMINERALISATION & DESALTING

Anion exchange
Reaction

Reverse osmosis &

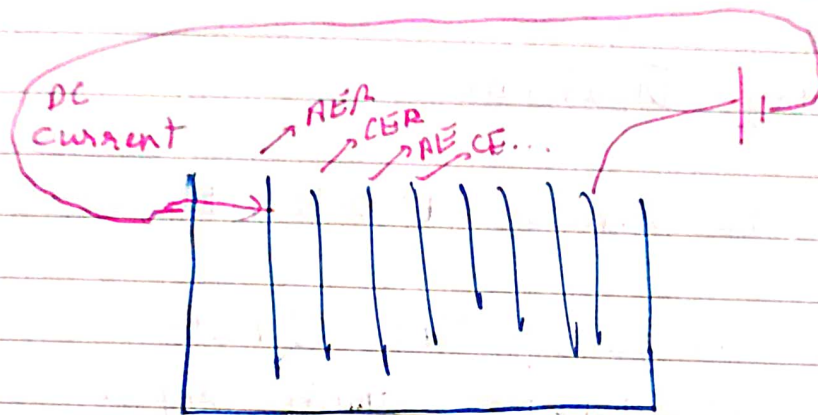
Electrodialysis

You can also achieve

some ~~deionisation~~

demineralisation

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- DC current is passed in the exchanger, cation exchanger takes cations with it and anion exchanger takes anion with it. There are multiple alternate anion and cation membranes placed next to each other.



- For R.O. :- Membrane separation process takes place, you apply external pressure, higher than osmotic pressure.

* MUNICIPAL WASTE WATER TREATMENT

Quality of Municipal Waste water:

a) Amount of Suspended solids.

b) Biochemical oxygen demand (BOD)

↙
Oxygen required to sustain microbes in water.

↘ More BOD ⇒ More sludge

Primary Treatment :-

- 1) Removal of suspended solids (30-80%) and reduce B.O.D.

- 2) Screening the large solids
- 3) Removal of settable solids by agglomeration and coagulation. (CLSP)
↳ Lime Soda

Secondary Treatment

- 1) Oxidation to remove B.O.D to 85-90%.
- 2) Activated Sludge system (Activated Sludge contains aerobic microorganisms which digest sludge).

Tertiary Treatment

- i) Phosphorous, nitrogen and Carbon ~~chemical~~ compounds are removed by chemical treatments

↓
precipitation with lime or metallic hydroxides (eg $Al(OH)_3$)

— x —

modified with phosphene.
stainless steel

Oxo process:- In a stirred tank reactor, made of stainless steel, Reactants and catalysts are fed from the bottom of the reactor. Catalyst → Rhodium catalyst → $Rh(CO)_5$. Due to the sensitivity to towards catalyst poisons, the syn gas must be carefully treated. The reactor contains the catalyst dissolved in reaction medium. A portion of the soln must be taken out, purified and again put into the reactor.

Reaction is exothermic. After 30% conversion, equilibrium is set, at a fixed T and P. Then you pass it through a dehumidifier and condenser. Unreacted reactants are reprocessed and recycled.
Along with ~~propane~~ propane

Product is freed from residual olefins and sent to the multistage distillation column to produce crude alcohol.